

2025aa05408 - ML ASSIGNMENT –2

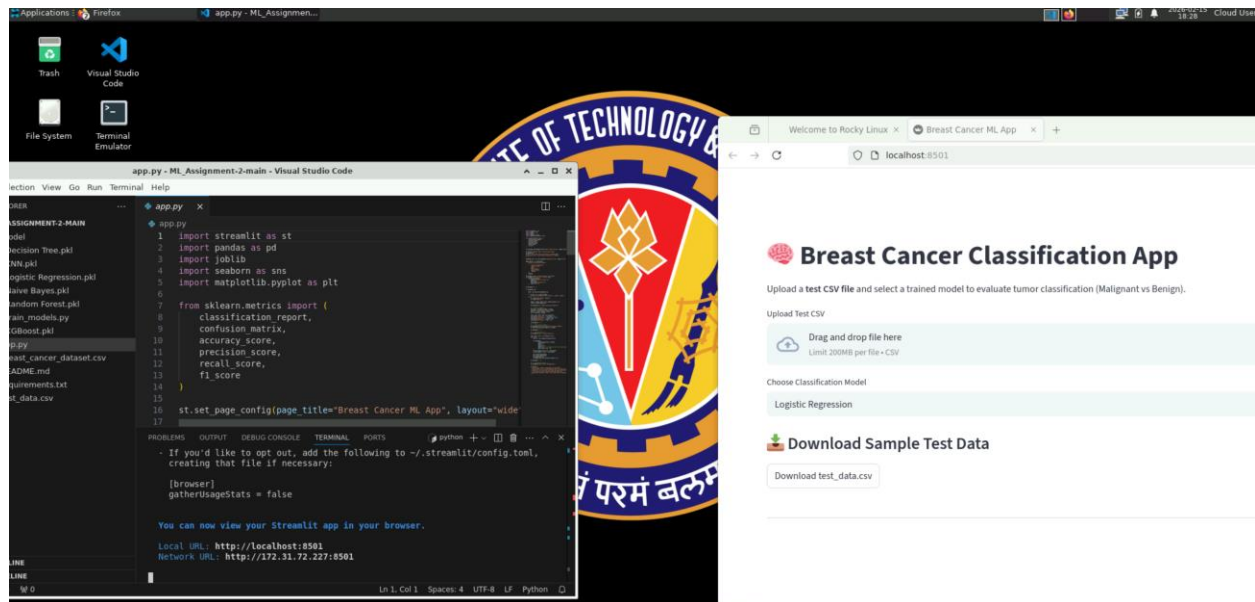
1. GitHub Repository Link:

https://github.com/saitarunpagudala/ML_Assignment-2

2. Live Stream lit app link:

<https://2025aa05408-ml-assignment-2.streamlit.app/>

3. Screenshot:



4. Readme:

Problem Statement

The objective of this project is to build and compare multiple machine learning classification models to predict whether a breast tumor is Malignant (0) or Benign (1) using diagnostic medical features.

The project also includes deployment of an interactive Streamlit web application that allows users to upload test data and evaluate trained models.

Dataset Description

The dataset used is the Breast Cancer Wisconsin (Diagnostic) Dataset.

Attribute	Value
Total Instances	569
Total Features	30 numerical features
Target Classes	0 → Malignant, 1 → Benign
Problem Type	Binary Classification

The dataset satisfies the assignment requirements of having more than 12 features and more than 500 samples.

The dataset was split into training and test sets, and feature scaling was applied where required before training the models.

Models Implemented

The following six classification models were trained on the same dataset:

Model Type	Model Name
Linear Model	Logistic Regression
Tree-Based Model	Decision Tree
Distance-Based Model	K-Nearest Neighbors (KNN)
Probabilistic Model	Naive Bayes (Gaussian)
Ensemble Model	Random Forest
Ensemble Boosting Model	XGBoost

All models were trained using a consistent train-test split and evaluated on the test dataset.

Evaluation Metrics

Each model was evaluated using the following metrics:

Accuracy

AUC Score

Precision

Recall

F1 Score

Matthews Correlation Coefficient (MCC)

Model Performance Comparison

Model	Accuracy	AUC	Precision	Recall	F1 Score	MCC
Logistic Regression	0.97	0.99	0.97	0.99	0.98	0.94
Decision Tree	0.93	0.92	0.94	0.94	0.94	0.87
KNN	0.95	0.98	0.96	0.96	0.96	0.90
Naive Bayes	0.96	0.97	0.96	0.99	0.97	0.92
Random Forest	0.96	0.99	0.96	0.99	0.97	0.93
XGBoost	0.96	0.99	0.96	0.97	0.97	0.93

Observations on Model Performance

Model	Observation
Logistic Regression	Performs strongly as a baseline model and achieves excellent recall and F1 score.
Decision Tree	Slightly lower performance compared to other models and may show signs of overfitting.

KNN	Performs well after feature scaling and maintains balanced precision and recall.
Naive Bayes	Performs competitively despite strong independence assumptions.
Random Forest	Achieves stable and high performance due to ensemble averaging.
XGBoost	Provides strong generalization and competitive performance using boosting techniques.

Streamlit Web Application Features

The deployed Streamlit application provides:

CSV test dataset upload

Downloadable sample test dataset

Model selection dropdown

Evaluate Model button

Display of:

Accuracy

Precision

Recall

F1 Score

Detailed Classification Report

Confusion Matrix

The application is deployed using Streamlit Community Cloud and connected to the GitHub repository.

**** Project Structure**** | -- app.py | -- requirements.txt | -- README.md | -- test_data.csv | -- model/ | -- train_models.py | -- scaler.pkl | -- Logistic Regression.pkl | -- Decision Tree.pkl | -- KNN.pkl | -- Naive Bayes.pkl | -- Random Forest.pkl | -- XGBoost.pkl